ChatGPT

Optimal Chemo Treatment Times

Computing the dosage rate a_2

The code sums terms of the form $(1/2)^k$ minus negligibly small $72^{-N k}$ terms. In fact one finds $$f(n)=\sum_{k=3}^{n+2}(1/2)^k=\frac{(1/2)^3\bigl(1-(1/2)^n\bigr)}{1-1/2}=\frac14\bigl(1-(1/2)^n\bigr)$$ by the standard geometric-series formula [1]. Hence $$a_2(n)=2f(n)+\frac12+(1/4)^n \;=\;1-\frac{(1/2)^{n+1}}{}+(1/4)^n+\text{(tiny terms)}\,, $$ which approaches 1 as $n\to\infty$. In other words the given computable definition yields $a_2\approx1$ (just slightly below 1 for any finite n).

Minimising the total time

We minimise x_1+x_2 subject to the dose constraint $a_1x_1 + a_2x_2 = 1$ (with $a_1=1$ and $a_2\approx1$). Since $a_1=1>a_2$, the first therapy is slightly faster. Using only the faster rate ($x_2=0$, $x_1=1$) achieves the required dose 1 in the least time. Numerically, to two-decimal accuracy one finds

- $x_1 = 1.00$, $x_2 = 0.00$,

which indeed gives total dose $1\cdot1.00 + a_2\cdot0.00 =1$ and minimises $x_1+x_2=1.00$. Thus the optimal schedule is to use only the first chemo for 1.00 s and skip the second.

Answer: $x_1=1.00,\;x_2=0.00$, with explanation as above.

Sources: Geometric series sum formula [1].

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[1] Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series